

1. Name of the Academic Unit: Civil Engineering

2. Subject Name: Water Resources and Geotechnical Engineering Sessional **L-T-P:** 0-0-3
Credits: 3

3. Pre-requisites: Hydraulics and Geotechnical Engineering

4. Syllabus and reference books:

Syllabus:

Vector and Raster Data Representation in standard GIS software; Rainfall data sources, interpretation, contouring; Digital Elevation Model (DEM); Watershed Delineation. 1D Channel routing, Data preparation, Simulation using a standard software. Hydraulic Model setup, Hydraulic Jump simulation using a standard CFD software. Design of Breakwater. Shallow foundation design. Deep foundation design: pile foundation. Earth retaining structures, embankment dams including seepage analysis. Slope stability.

Reference Books:

1. Hydraulics Engineering Softwares
2. REEF3D or OpenFoam
3. Geotechnical Engineering Softwares

5. Practical-wise break-up:

Sl. No.	Topic	No. of Practical Hours
1.	Vector and Raster Data Representation in standard GIS software; Rainfall data sources, interpretation, contouring; Digital Elevation Model (DEM); Watershed Delineation	3
2.	1D Channel routing, Data preparation, Simulation using a standard software	6
3.	Hydraulic Model setup, Hydraulic Jump simulation using a standard CFD software	6
4.	Shallow foundation design	6
5.	Deep foundation design: pile foundation	3
6.	Earth retaining structures, embankment dams including seepage analysis	6
7.	Slope stability	3
8.	Design of Breakwater	3
Total number of hours		36